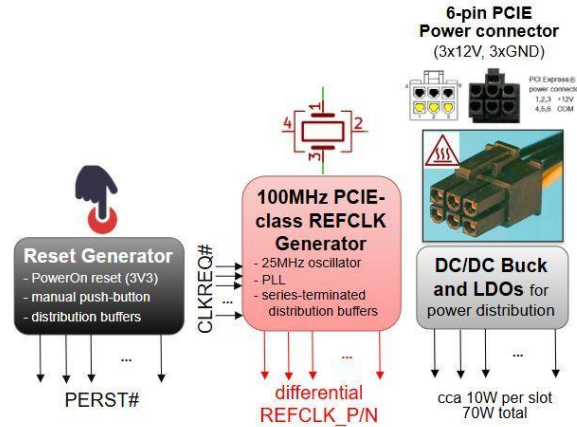


The background is a vibrant blue with a complex pattern of white and light blue lines and circles, resembling a circuit board or a network diagram. A large, solid black rectangle is centered in the middle of the image, serving as a backdrop for the text.

OPENPCIE

BACKPLANE

--- openPCIE backplane ---



4-lane "Direct" island RC4 => EP4

RC4

4-lane RC connector with mechanical option:
- M.2 (PCIe)

4 lanes

EP4

4-lane EP connector with mechanical option:
- Slot

1-lane "Switched" island RC1=>SW=>SW_EP0/1/2/3

RC1

1-lane RC connector with mechanical option:
- Slot

1 lane

Switch
PCIe2.0
1-lane
1-to-4

4 x
1 lane

1-lane EP connector - Slot
SW_EP0
1-lane EP connector - Slot
SW_EP1
1-lane EP connector - M.2 (PCIe)
SW_EP2
1-lane EP connector - M.2 (PCIe)
SW_EP3

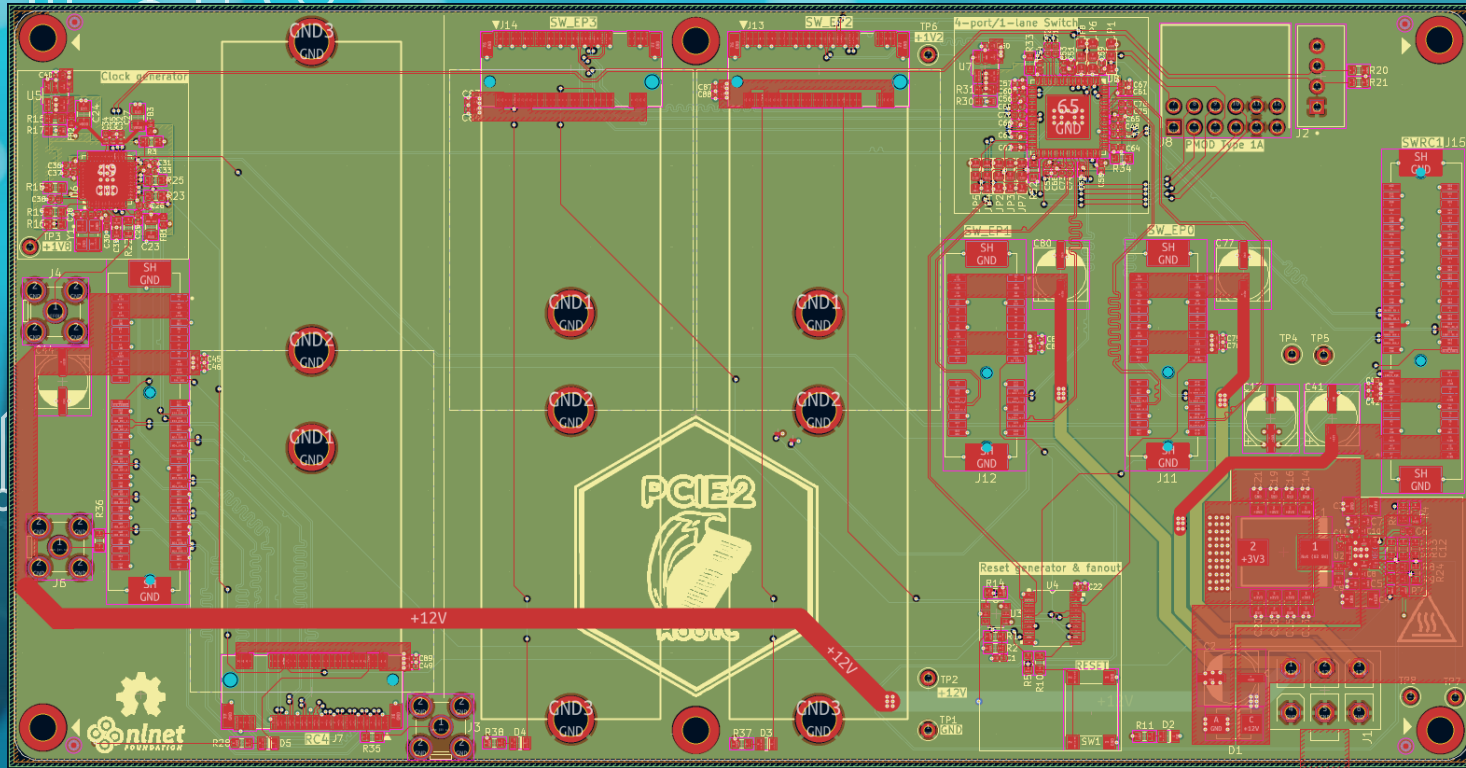
The "RC Connectors" are for the plug-in cards that are natively EP.

The backplane job is to swap the connector pins so that the Tx diff.pair on one side of the link is connected to the Rx diff.pair on the other side, and vice-versa. This allows the same FPGA plug-in card, which is always pinned-out to serve as an EP, to be used in both EP and RC roles.

FUNCTIONALITY AND TOPOLOGY

- Two Logical Islands:
 - 4-lane "Direct" connection
 - 1-lane "Switched" connection
- Supports standard PCIe Slots and M.2 connectors
- Single 6-pin PCIe power connector
- Integrated 100 MHz clock and reset distribution
- Same FPGA plug-in card can act as EndPoint (EP) or Root Complex (RC)

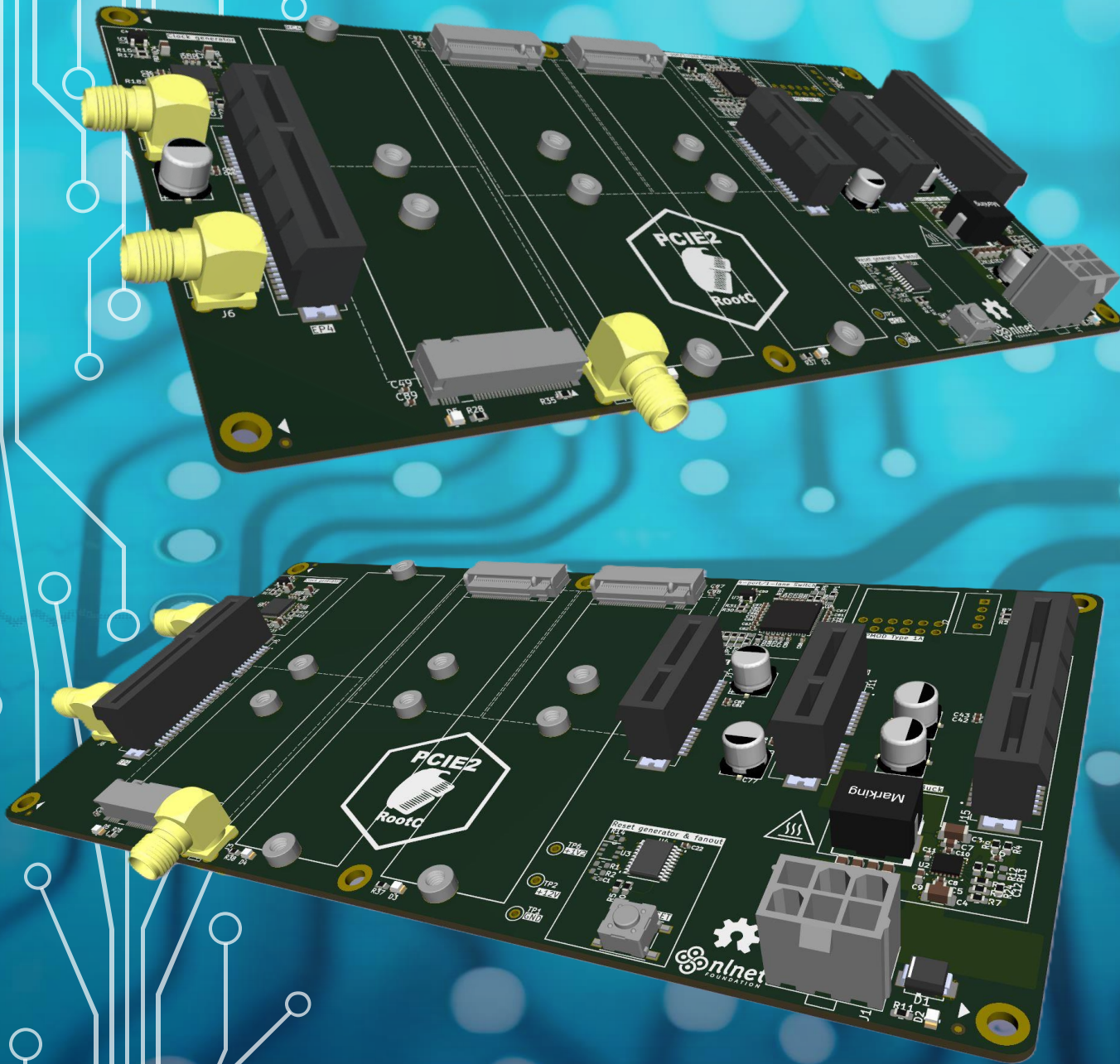
KICAD PCB LAYOUT



- Designed in KiCad
- 4-Layer Stackup:
 - Top: Microstrip for diff pairs
 - L2: Ground plane
 - L3: 3.3V Power plane
 - Bottom: Microstrip for diff pairs
- Careful routing to minimize stubs and skew

3D VIEW

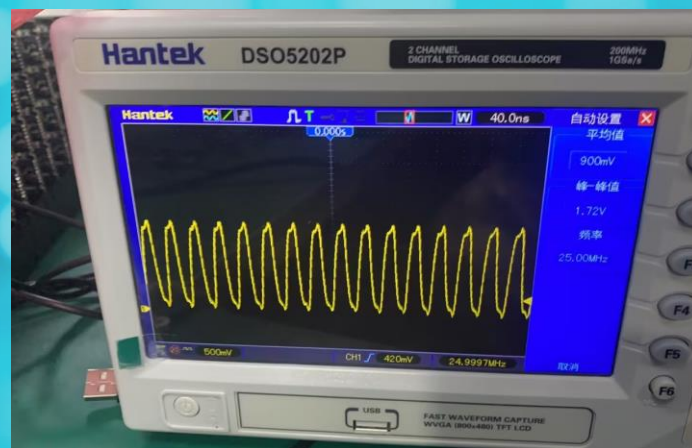
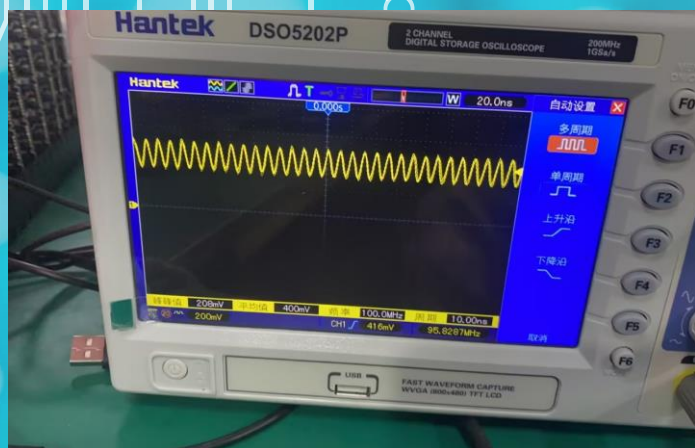
- 3D visualization of the openPCIE backplane



MANUFACTURING AND TESTING PREPARATION

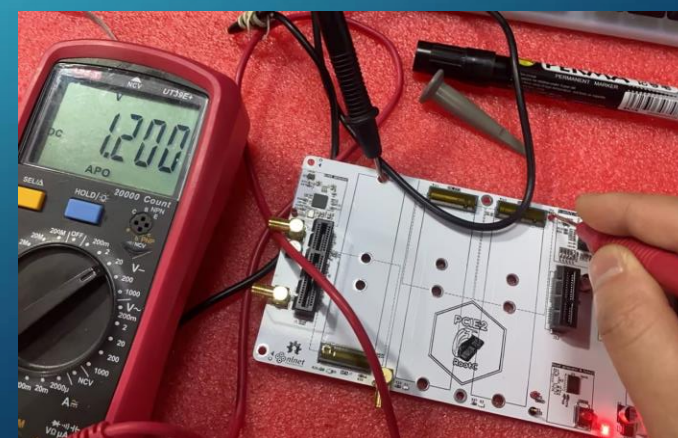
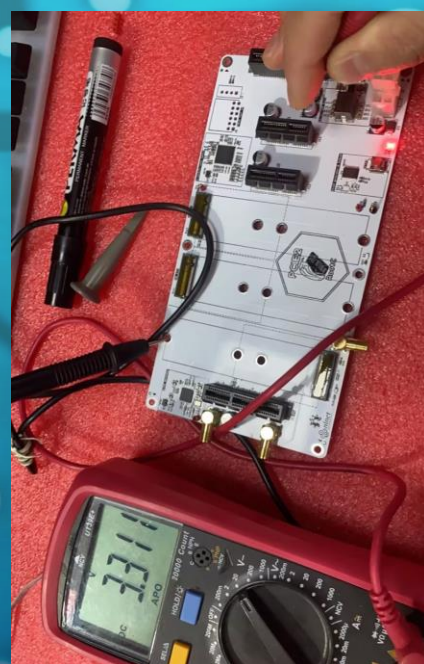
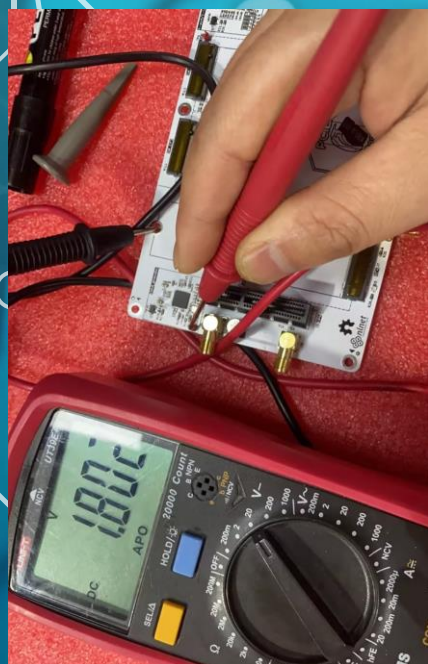


- PCB manufacturing by Elecrow (5 prototypes)
- Created a PCBA Functional Test Procedure
- Testing basic power, reset, and clock generation



FACTORY TEST RESULTS

- Factory tests passed successfully
- Clock (25MHz, 100MHz) and voltages (1.8V, 12V, 3.3V, 1.2V) verified





BOARDS ARRIVAL

- Prototypes arrived in Sarajevo
- Ready for hardware validation

STARTING THE TESTS (DIRECT CONNECTION)

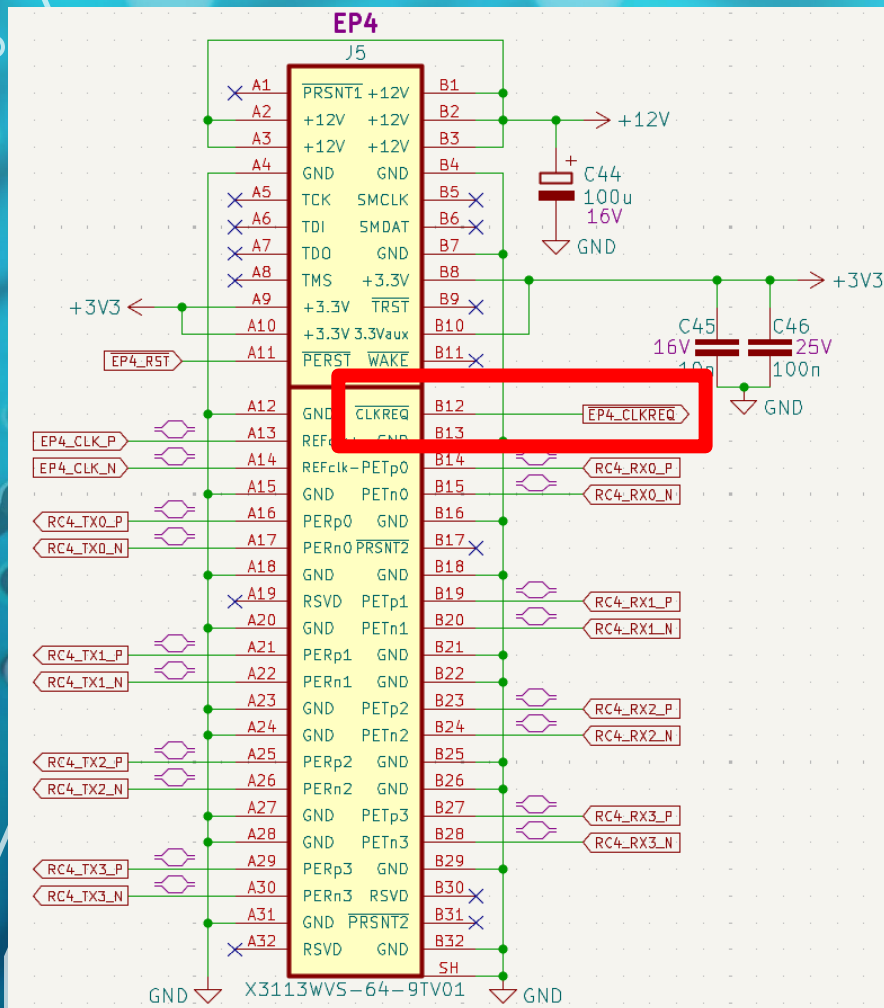
- Phase 1: Testing the Direct Connection (RC to EP)
- Inserting the FPGA cards





THE CLKREQ PROBLEM

- The Problem: CLKREQ# signal
- Clock generator requires CLKREQ to be pulled to GND
- Assumption: most M.2 to PCIe adapters route this signal to the M.2 connector, allowing the FPGA to pull it to Ground.



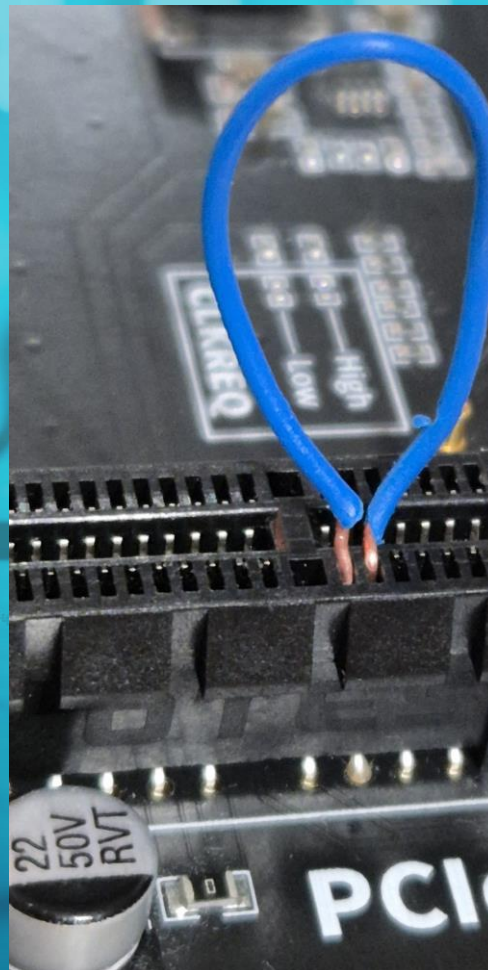
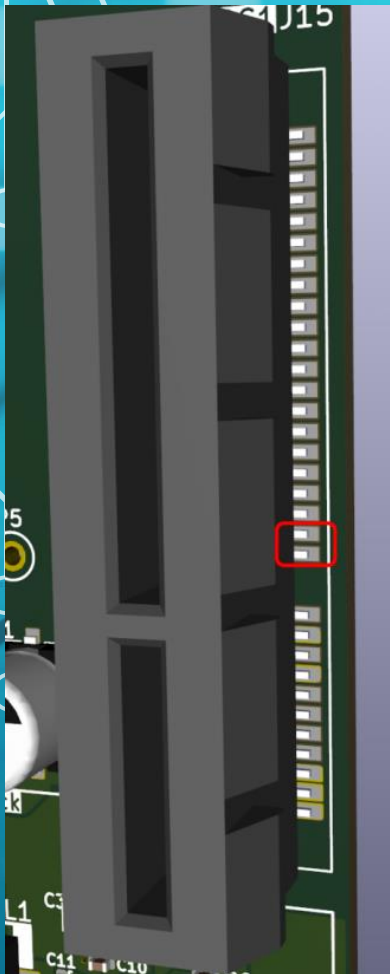
Pin#		Side B	Side A	
	Name	Description	Name	Description
1	+12V	12 V power	PRSNT1#	Hot-Plug presence detect
2	+12V	12 V power	+12V	12 V power
3	+12V	12 V power	+12V	12 V power
4	GND	Ground	GND	Ground
5	SMCLK	SMBus (System Management Bus) clock	JTAG2	TCK (Test Clock), clock input for JTAG interface
6	SMDAT	SMBus (System Management Bus) data	JTAG3	TDI (Test Data Input)
7	GND	Ground	JTAG4	TDO (Test Data Output)
8	+3.3V	3.3 V power	JTAG5	TMS (Test Mode Select)
9	JTAG1	TRST# (Test Reset) resets the JTAG interface	+3.3V	3.3 V power
10	3.3Vaux	3.3 V auxiliary power	+3.3V	3.3 V power
11	WAKE#	Signal for Link reactivation	PERST#	Fundamental reset
Mechanical key				
12	RSVD	Reserved	GND	Ground
13	GND	Ground	REFCLK+	Reference clock (differential pair)
14	PETp0	Transmitter differential pair, Lane 0	REFCLK-	
15	PETn0	Lane 0	GND	Ground
16	GND	Ground	PERp0	Receiver differential pair, Lane 0
17	PRSNT2#	Hot-Plug presence detect	PERn0	
18	GND	Ground	GND	Ground
End of the x1 connector				
19	PETp1	Transmitter differential pair, Lane 1	RSVD	
20	PETn1	Lane 1	GND	Ground

Table 37: PCI Express Connectors Pinout

Pin #	Name	Side B Description	Name	Side A Description
1	+12V	+12 V power	PRSNT1#	Hot-Plug presence detect
2	+12V	+12 V power	+12V	+12 V power
3	+12V	+12 V power	+12V	+12 V power
4	GND	Ground	GND	Ground
5	SMBCLK	SMBus (System Management Bus) clock	JTAG2	TCK (Test Clock), clock input for JTAG interface
6	SMBDAT	SMBus (System Management Bus) data	JTAG3	TDI (Test Data Input)
7	GND	Ground	JTAG4	TDO (Test Data Output)
8	+3.3V	+3.3 V power	JTAG5	TMS (Test Mode Select)
9	JTAG1	TRST# (Test Reset) resets the JTAG interface	+3.3V	+3.3 V power
10	+3.3Vaux	+3.3 V auxiliary power	+3.3V	+3.3 V power
11	WAKE#	Signal for Link reactivation	PERST#	Fundamental reset
Mechanical Key				
12	CLKREQ#	Clock Request Signal	GND	Ground
13	GND	Ground	REFCLK+	Reference clock (differential pair)
14	PETp0	Transmitter differential pair, Lane 0	REFCLK-	
15	PETn0	Transmitter differential pair, Lane 0	GND	Ground
16	GND	Ground	PERp0	Receiver differential pair, Lane 0
17	PRSNT2#	Hot-Plug presence detect	PERn0	
18	GND	Ground	GND	Ground
End of the x1 Connector				
19	PETp1	Transmitter differential pair, Lane 1	RSVD	
20	PETn1		GND	Ground
21	GND	Ground	PERp1	Receiver differential pair, Lane 1
22	GND	Ground	PERn1	
23	PETp2	Transmitter differential pair, Lane 2	GND	Ground

HISTORY OF CLKREQ

- CLKREQ# was introduced in Revision 4 of the PCIe electromechanical specification in late 2018
- Previously, pin B12 was marked as "Reserved"
- Older adapters do not connect this pin



TEMPORARY FIX

- First Attempt: put a simple wire between CLKREQ and Ground.
- Issue: Created an inductive loop/antenna near tx0
- Resulted in an unstable link on Lane 0



THE PERMANENT FIX

- Better Solution: Solder bridge between CLKREQ and GND
- Clock is only active when needed
- Pin layout is perfect for a simple solder bridge:
 - The Ground pin is right next to the CLKREQ pin
 - The CLKREQ is the last pin on the left side of the connector

VERIFICATION ENVIRONMENT

- Dual-PC Setup:
 - PC 1: Connected to RC FPGA via JTAG
 - PC 2: Connected to EP FPGA via JTAG
- Speed and Efficiency: No manual swapping of JTAG cables and no time-consuming flashing
- Simultaneous Debugging: Run two Vivado Logic Analyzers at the same time





TEST OBJECTIVE AND RESULTS

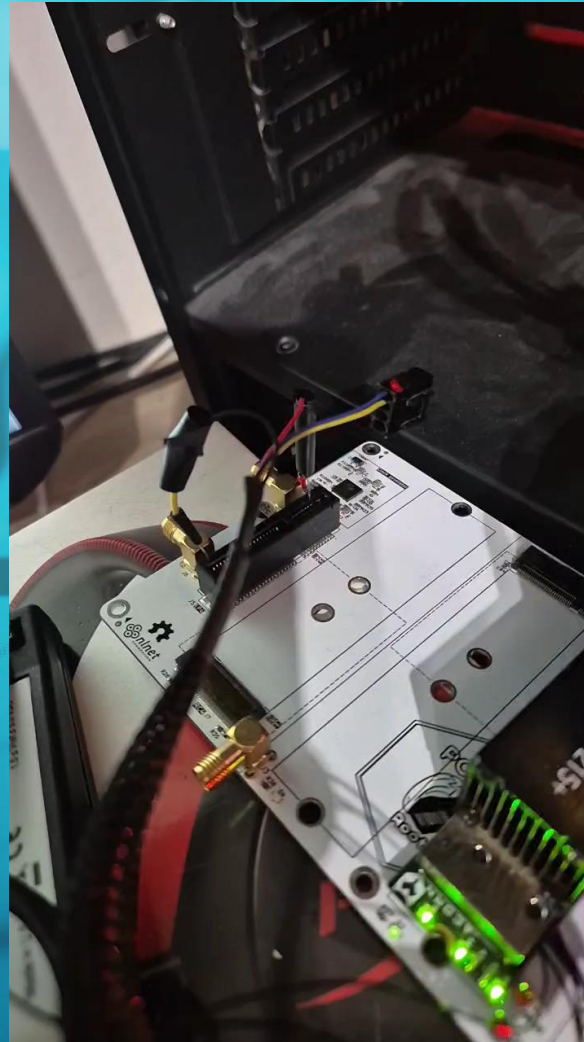
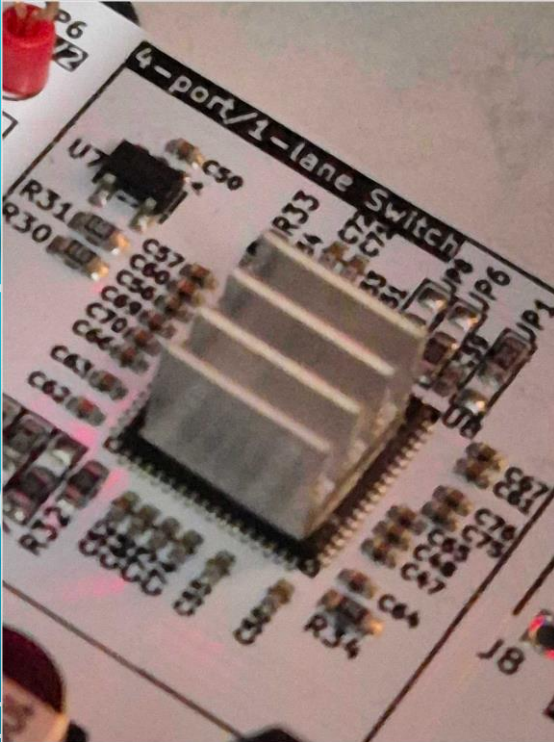
- Objective: Verify Link Enumeration, Memory Write, and Memory Read
 - Write: RC sends a data payload (Decimal 6) to EP
 - Transfer: EP writes it into internal Block RAM
 - Read: RC reads the data back
- Result: Successful link (LED is ON) and data matches (0110 binary)



SWITCH TESTING

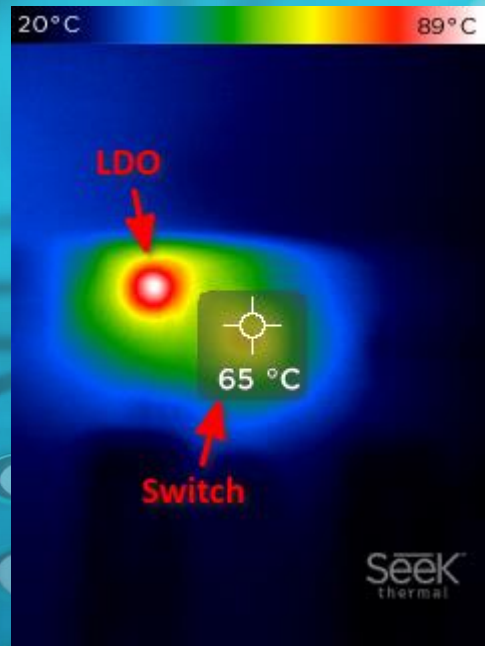
- Moving from Direct connection to the PCIe Switch
- Inserting boards into the 1-to-4 Switch topology

THE SWITCH OVERHEATING PROBLEM



- New Problem: One area of the board board was getting extremely hot.
- Switch was too hot to touch; link was highly unstable
- Temporary fix: Added a heatsink and fan
- Investigation: Thermal camera revealed the real issue (1.2V LDO)

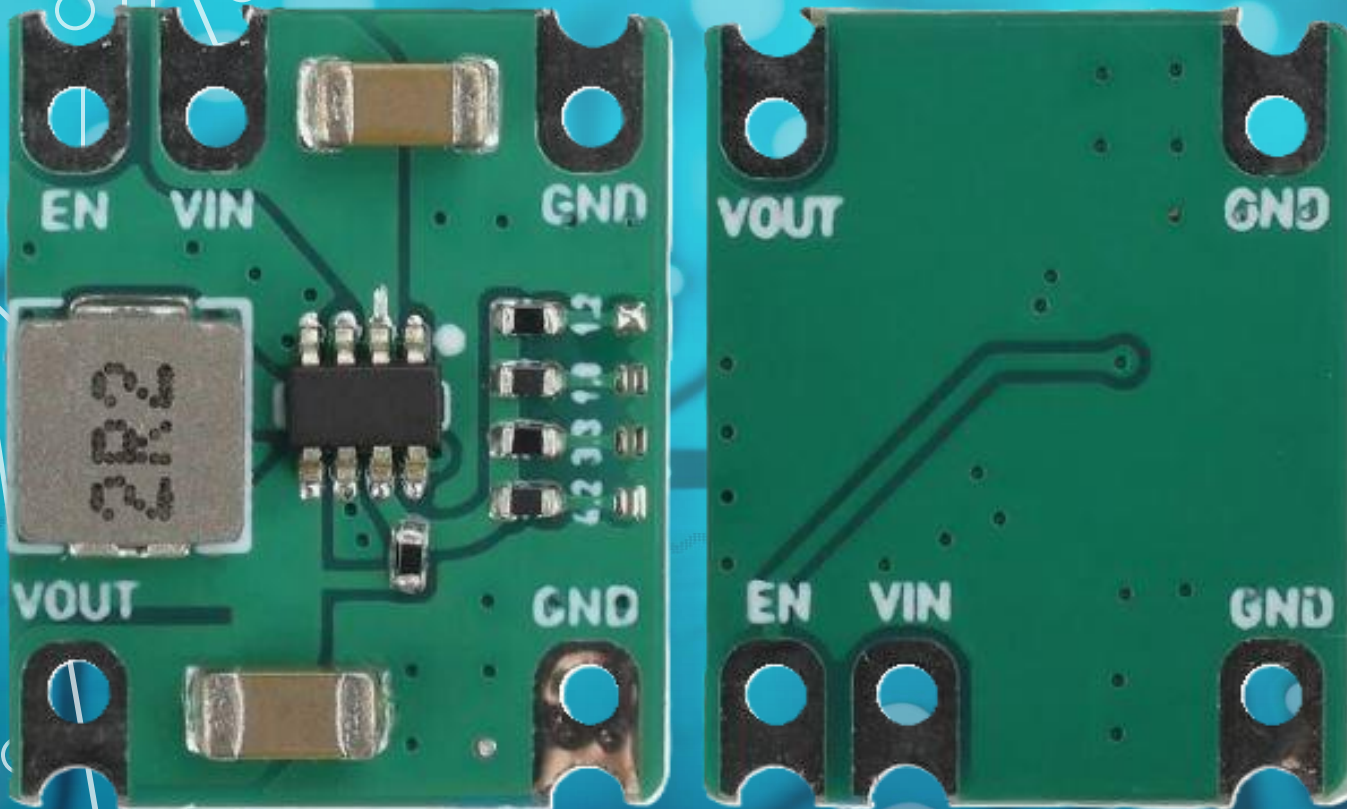
THERMAL CAMERA ANALYSIS



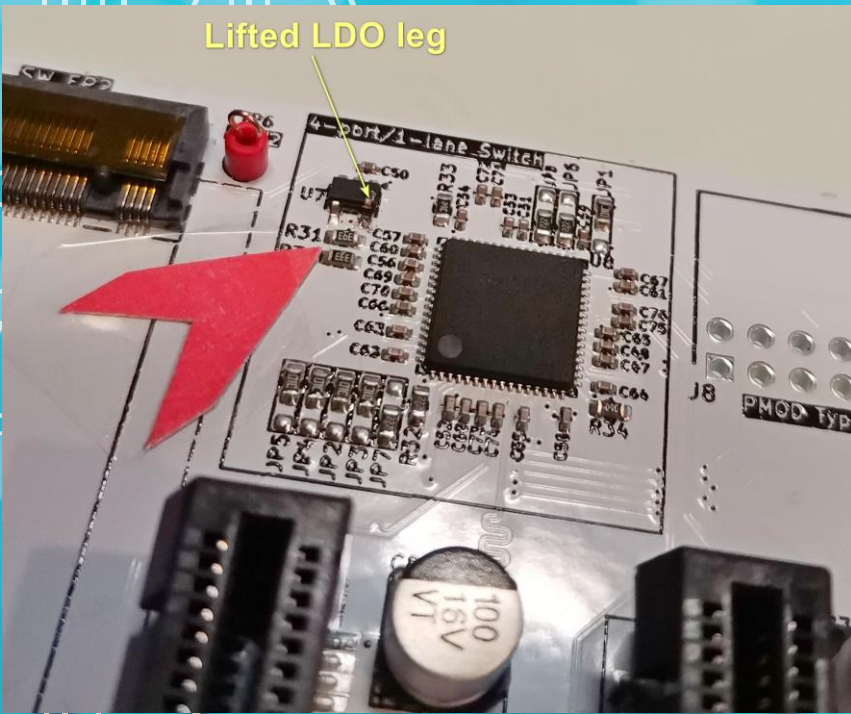
- Thermal camera shows LDO at 90°C
- Switch pulls up to 2 Watts
- LDO cannot handle this power dissipation, so it was overheating.

TESTING THE BUCK CONVERTER

- Ordered a DC-DC Step-Down Power Module
- Input: 2.5-6V | Output: 1.2V Voltage Regulator
- Manual installation for testing

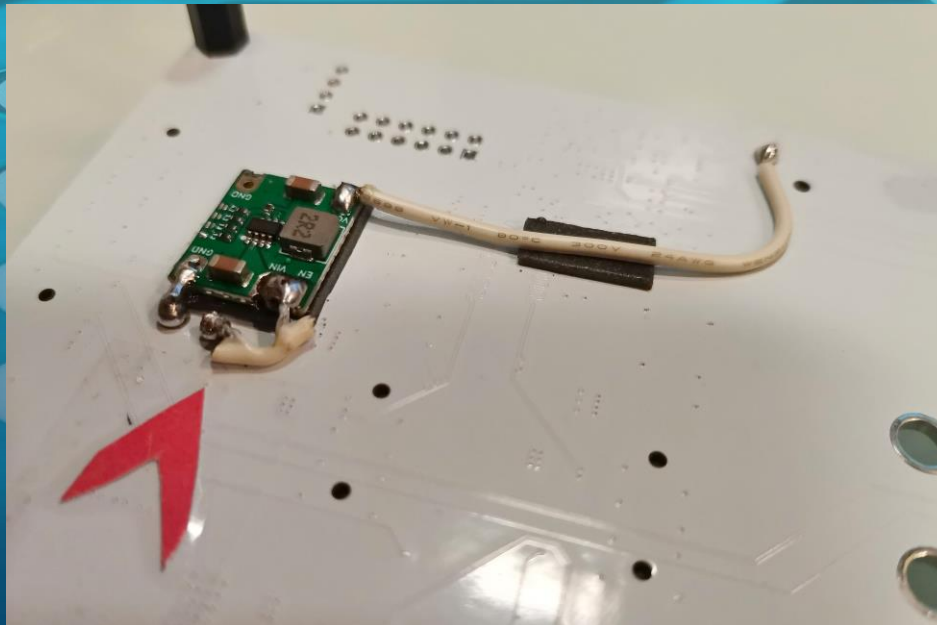


Lifted LDO leg

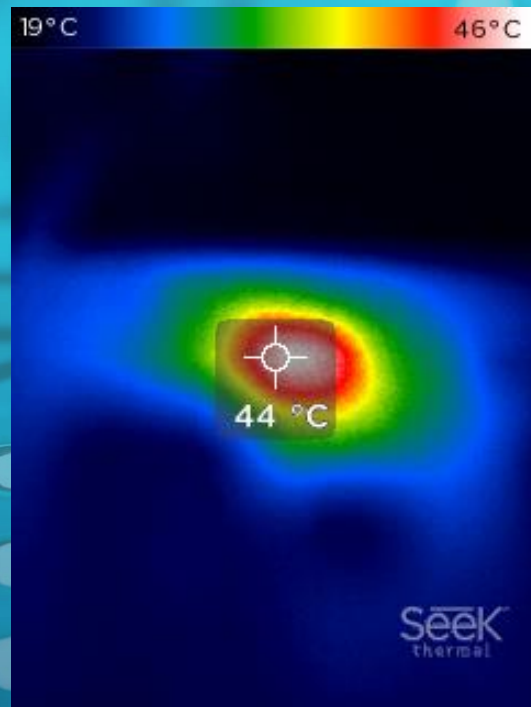


MANUAL BOARD MODIFICATION

- Lifted the LDO leg to disconnect it
- Wired the new DC-DC Buck converter to the board



TEMPERATURE RESULTS



- Temperature dropped to $\sim 45^{\circ}\text{C}$
- Stable communication restored
- Second batch: LDO will be replaced by a Buck converter



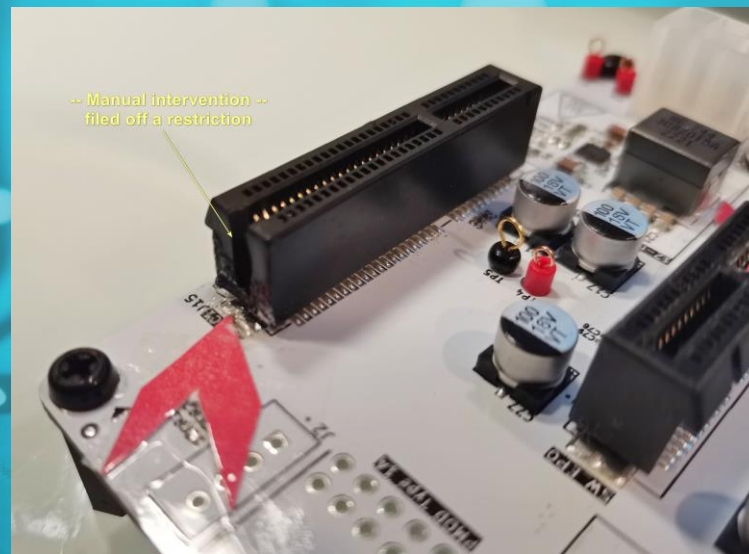
FULLY POPULATED SWITCH

- Fully populated PCIe switch setup

Open end allows
plugging-in card
of any length.



-- Manual intervention --
filed off a restriction

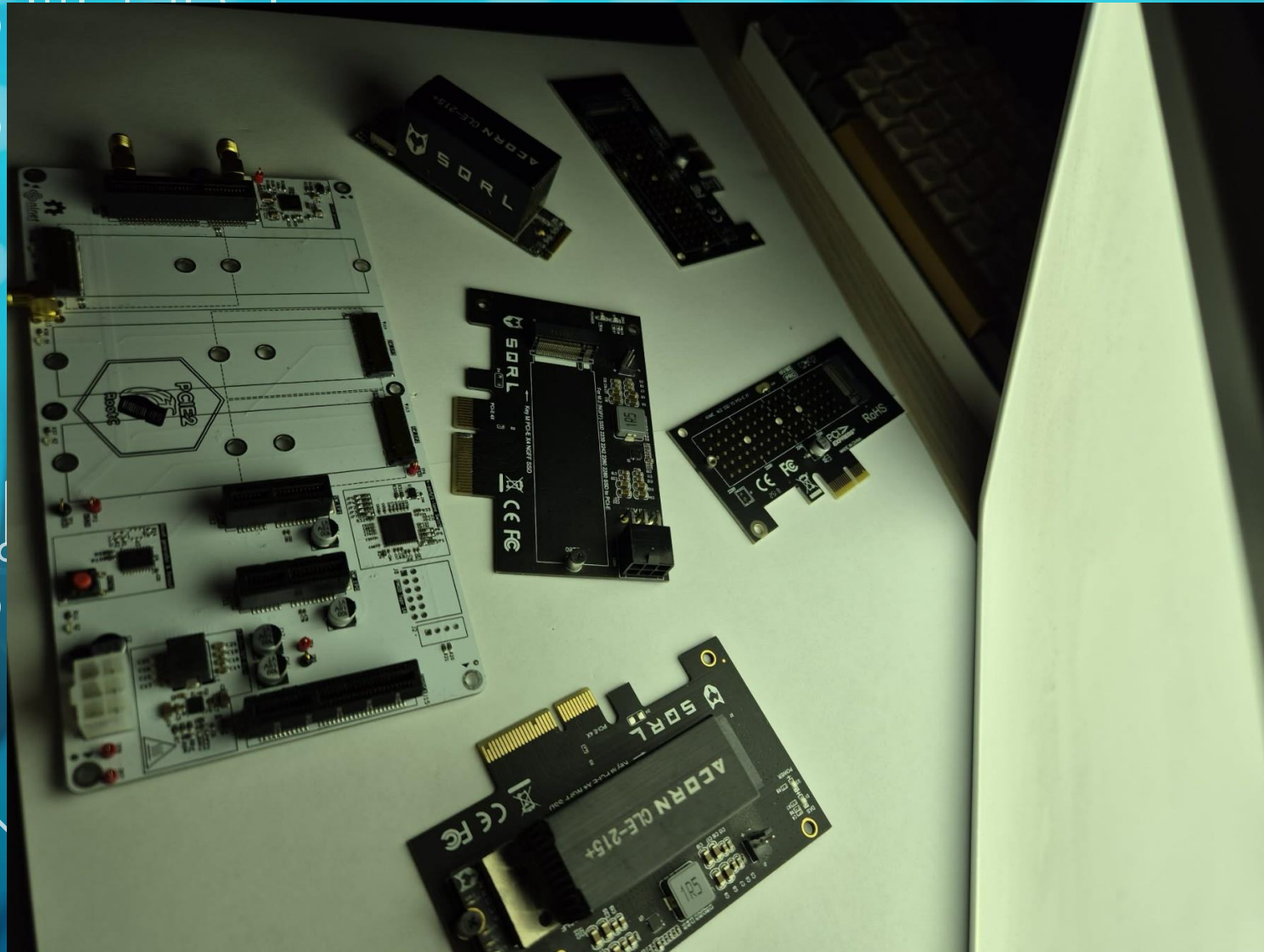


- Manual intervention -
filed off a restriction



IMPROVEMENTS FOR THE SECOND BATCH

- Updates for the next PCB revision:
 - Open-end PCIe slots (allows x4 cards in x1 slots)
 - Add power switch for the main 12V supply line
 - Replace LDO with DC-DC Buck converter
 - Add CLKREQ shunt on the backplane
 - Other minor improvements



THANK
YOU FOR
YOUR
ATTENTION!